Dual-Encoder versus Cross-Encoder Neural Architectures for Gene Regulatory Network Inference: An Empirical Comparison Under Parameter-Matched Conditions

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Abstract—Gene regulatory network (GRN) inference—determining which transcription factors regulate which target genes from single-cell RNA-sequencing data—is a core problem in computational biology. We present a parameter-matched comparison of dual-encoder (two-tower) and cross-encoder MLPs on a human brain single-nucleus RNA-seq dataset (39K TF–gene pairs, 5.58M parameters each) under balanced (1:1) and moderately imbalanced (5:1) negative-to-positive sampling ratios. The cross-encoder achieves AUROC 0.9025 versus 0.7941 for the two-tower under balanced training—a gap of 10.8 points consistent across five random seeds ($t(4) = 27.08$, $p < 0.0001$, Bonferroni-corrected $\alpha = 0.017$, Cohen's $d = 12.1$). A no-product cross-encoder ablation attributes ≈95% of this gap to joint encoding and ≈5% to the multiplicative interaction term. Under 5:1 negative sampling, the two-tower AUROC degrades by 10.1 points while the cross-encoder remains stable; both architectures exhibit F1 degradation. Structured neuron pruning across three random seeds shows the two-tower maintains post-hoc AUROC retention ≥ 0.997 up to 60% sparsity, with mean retention falling to 0.978 at 90% sparsity; the cross-encoder shows comparable post-hoc stability through 70% sparsity. Both architectures operate in a transductive setting (entity embeddings are learned from the training split and cannot generalize to unseen TFs or genes), and the evaluation task measures reconstruction of curated regulatory databases rather than inference of novel regulation. All findings are on a single human brain dataset and are qualified to that experimental context; comparison to established GRN inference baselines (GENIE3, SCENIC) is a necessary next step.

Index Terms—gene regulatory network inference, dual-encoder, two-tower architecture, cross-encoder, neural network pruning, single-cell RNA-seq, representation learning, link prediction, class imbalance, transductive learning

I. Introduction

Gene regulatory network (GRN) inference — the problem of identifying which transcription factors (TFs) regulate which target genes — is a fundamental challenge in computational biology. Understanding transcriptional regulatory relationships is essential for deciphering developmental programs, disease mechanisms, and therapeutic targets. With the availability of large-scale single-nucleus RNA-sequencing (snRNA-seq) data, machine learning methods have emerged that cast GRN inference as binary link prediction: given a candidate TF–gene pair and their expression profiles, predict whether a regulatory relationship exists [1], [2].

A central architectural question in link prediction is whether entity representations should be computed independently (a dual-encoder or two-tower model [3], [4]) or jointly (a cross-encoder [5], [6]). Two-tower models score interactions via cosine similarity of independently computed encodings, enabling efficient inference at scale. Cross-encoders process pairs jointly, allowing arbitrary interaction features to be learned before scoring. In information retrieval, cross-encoders consistently outperform two-tower models when precise ranking quality is needed [5], [6], but whether this advantage holds in computational biology—where entities have distinct biological identities and the regulatory signal is low-dimensional—has not been established under controlled, parameter-matched conditions.

Three open questions motivate this work: (1) Does a cross-encoder achieve higher discriminative AUROC than a parameter-matched two-tower for TF–gene link prediction? (2) Is the two-tower's cosine similarity scoring more sensitive to class imbalance than the cross-encoder's MLP scoring? (3) Does the trained two-tower actually utilize its nominal hidden-layer capacity, or is much of it redundant?

This paper makes three contributions:

- C1: A parameter-matched comparison (5.58M parameters each) of two-tower and cross-encoder MLPs on human brain snRNA-seq data under both balanced (1:1) and moderately imbalanced (5:1) negative-to-positive sampling ratios, including an ablation of the cross-encoder without its element-wise interaction term. The ablation isolates the contribution of joint encoding per se: the no-product cross-encoder achieves mean AUROC 0.8972 (1:1) versus 0.9025 for the full cross-encoder and 0.7941 for the two-tower, attributing ≈95% of the 10.8-point gap to joint encoding and ≈5% to the element-wise product term (see Section IV-C).
- C2: Demonstration that the two-tower degrades by 10.1 AUROC points under moderately imbalanced (5:1) negative sampling ($t(4.25) = 4.06$, $p = 0.014$) while the cross-encoder AUROC is stable (F1 degrades for both architectures); the true TF–gene

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imbalance is ≈376:1 and constitutes an open question.

- C3: A structured neuron pruning analysis across three random seeds showing that post-hoc AUROC retention is ≥ 0.997 (mean across seeds) up to 60% sparsity for the two-tower, declining to 0.978 at 90% sparsity. The same protocol applied to the cross-encoder shows post-hoc retention ≥ 0.999 through 70% sparsity, dropping to 0.987 at 90%, demonstrating that representational redundancy is not exclusive to cosine similarity scoring.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset, architectures, training protocol, ablation design, and pruning procedure. Section IV presents experimental results, including the element-wise product ablation, cross-encoder pruning, and temperature sensitivity analysis. Section IV-G discusses implications and limitations. Section V concludes.

II. Related Work

A. Gene Regulatory Network Inference as Link Prediction

Gene regulatory network inference is the problem of identifying directed edges in a graph where nodes are genes and edges represent transcriptional regulation. Curated databases such as DoRothEA [7] and TRRUST [8] provide partial ground truth derived from ChIP-seq experiments and literature curation, but coverage is sparse relative to the space of possible regulatory interactions.

Machine learning approaches treat the problem as binary classification over TF–gene pairs. Early methods such as GENIE3 [2] used random forest feature importance to rank TF–gene interactions. SCENIC [1] extended this with cis-regulatory motif analysis. GRNBoost2 [9] improved upon GENIE3 with gradient boosting, achieving substantially faster runtimes with comparable accuracy. Information-theoretic methods such as PIDC [10] exploit multivariate mutual information to capture higher-order regulatory dependencies that pairwise correlation measures miss. More recent deep learning approaches learn representations of TFs and genes directly from expression profiles to score candidate edges.

The scale of the problem is biologically daunting: the human genome encodes approximately 1,600 transcription factors [11], each potentially regulating hundreds to thousands of target genes. Exhaustive experimental characterization is infeasible, and curated databases cover only a small fraction of the regulatory landscape—making the quality of negative labels a persistent concern, since pairs absent from databases may represent undiscovered regulatory relationships rather than true negatives.

B. Two-Tower (Dual-Encoder) Architectures

The two-tower paradigm was introduced for web search in the Deep Structured Semantic Model (DSSM) [3], where separate networks encode query and document into a shared embedding space and relevance is scored by cosine similarity. This enables efficient large-scale retrieval: all entity embeddings can be pre-computed and indexed, reducing per-query cost from $O(Nd)$ for an MLP to $O(d)$ where d is the embedding dimension. Dense Passage Retrieval (DPR) [12] demonstrated the practical power of this approach at scale: pre-computed passage embeddings indexed with approximate nearest-neighbor search enable millisecond-latency retrieval over millions of documents, a property highly relevant for genome-scale GRN inference where the TF×gene product space can reach 10^8 candidate pairs.

Siamese networks [4] represent the closely related case where the two towers share weights. In the GRN setting, TF and gene encoders are kept separate (non-Siamese) to allow distinct representation spaces for the two biological entity types.

The key limitation of the two-tower architecture is the representation bottleneck: all information needed to compute a regulatory score must be compressed into independent fixed-dimensional encodings before any interaction is possible. This precludes learning features that are inherently joint — for example, regulatory interactions predictable only from the simultaneous observation of TF expression level and gene chromatin accessibility.

C. Cross-Encoder Architectures

Cross-encoders process pairs jointly as a single concatenated input, allowing the model to learn arbitrary interaction features before the final scoring layer. Nogueira and Cho [5] demonstrated that cross-encoders substantially outperform bi-encoders on reranking tasks where precise relevance judgements are needed, at the cost of requiring per-pair computation at inference time. Humeau et al. [6] further characterized this trade-off in Poly-encoders, and Khattab and Zaharia [13] explored late interaction in ColBERT as a hybrid approach. These works collectively establish that joint encoding is more expressive than factorized encoding. In natural language processing, BERT [14] is the dominant cross-encoder backbone: its transformer self-attention mechanism allows every token in the query to attend to every token in the document before the scoring layer, achieving deep pairwise interaction at the cost of $O(N^2)$ per-pair computation. The MLP cross-encoder used in this work adopts the same inductive bias—joint input processing before scoring—without the quadratic attention cost, which is tractable for fixed-length embedding inputs.

The element-wise product $\mathbf{a} \odot \mathbf{b}$ of two embeddings is a standard feature engineering technique that provides a direct path for learning which embedding dimensions co-activate for relevant pairs [15].

D. Structured Pruning and Representational Redundancy

Neural network pruning removes parameters from a trained network to reduce model size or inference cost while preserving performance. Unstructured pruning removes individual weights [16]; structured pruning removes

entire neurons or filters, producing actual speedups without sparse-matrix libraries. The earliest principled pruning criterion, Optimal Brain Damage [17], used second-order Taylor expansions to identify and remove low-saliency weights; this saliency-based perspective underpins many modern importance scores including the activation-frequency and weight-magnitude criteria used in this work.

Han et al. [16] demonstrated that large networks tolerate removal of 80–90% of weights with minimal accuracy loss, suggesting that trained networks are highly over-parameterized. The Lottery Ticket Hypothesis [18] further showed that sparse sub-networks exist within trained networks that can be trained in isolation to match the full network, implying that useful representational capacity is localized. A comprehensive survey of pruning methodology [19] found that reported results across the literature are difficult to compare due to inconsistencies in baselines, fine-tuning budgets, and sparsity definitions; it recommends reporting both immediate post-pruning and fine-tuned performance at each sparsity level—a convention this paper follows.

III. Methods

A. Dataset and Prior Knowledge

Gene expression data were obtained from a human brain single-nucleus RNA-seq dataset (CellxGene Census accession fe1a73ab-a203-45fd-84e9-0f7fd19efcbd; human, Homo sapiens, brain tissue). Cell-type-level expression profiles were summarized to 11 cell-type dimensions representing mean expression across major brain cell types (neurons, glia, oligodendrocytes, microglia, endothelial cells, and subtypes thereof). Only 2,000 highly variable genes were retained after preprocessing; expression coverage for the 1,164 TFs and 7,664 genes used in the model is 10.7% and 14.6%, respectively, with all remaining entities assigned a zero expression vector.

Prior regulatory knowledge was sourced from merged DoRothEA [7] and TRRUST [8] databases, yielding approximately 29,000 TF–gene pairs as candidate positive examples. After filtering for TFs and genes with available expression data, approximately 23,700 positive pairs are retained. An equal number of negative (non-regulatory) examples were sampled uniformly from the space of TF–gene pairs not present in the prior databases, producing a balanced dataset of approximately 47,400 total examples. The dataset was split into training (70%), validation (15%), and test (15%) sets using a fixed random seed (42) applied once before any model training, ensuring that all model comparisons use identical data splits (test set: 7,109 examples). Note that the space of possible TF–gene pairs is $1{,}164 \times 7{,}664 \approx 8.9$ million; the true non-edge to edge ratio is therefore approximately 376:1, far more severe than any ratio tested here. The human genome encodes approximately 1,600 transcription factors in total [11]; the 1,164 TFs retained here represent those for which both prior-knowledge annotations and expression data are available in this dataset.

Transductive learning setting. Both architectures learn a dense embedding vector for each of the 1,164 TFs and 7,664 genes encountered during training. These embedding parameters (4.52M of the total 5.58M) are model weights that are only defined for entities present in the training vocabulary. The evaluation therefore operates in a transductive setting: test-set TF–gene pairs involve the same TFs and genes seen during training, and the model cannot produce predictions for entities outside its training vocabulary without retraining. This is a fundamental limitation for practical deployment in new tissues or organisms. Inductive extensions (e.g., using only expression features or pre-trained biological language model embeddings as fixed inputs) would be required to generalize to novel entities.

Task framing and label quality. Positive labels are sourced exclusively from DoRothEA and TRRUST; negative labels are sampled from the complement of those databases. The evaluation therefore measures how well a model reconstructs the curated database from expression features and learned entity embeddings, not whether it can discover novel regulatory relationships. Database coverage is sparse: only ≈0.27% of the 8.9 million possible TF–gene pairs are annotated as positive. Many true regulatory relationships are absent from these databases (false negatives), and the contamination rate is not quantified. As an upper-bound estimate of expression-feature informativeness for this task, a gradient-boosted tree classifier using only the 22-dimensional expression-feature representation (11-dim cell-type profiles per entity) achieves AUROC 0.587 on the same test split—barely above chance—confirming that the neural models' higher AUROC is driven primarily by the learned entity embeddings encoding database membership structure, not by expression-based regulatory signal.

B. Model Architectures

Two-Tower MLP. The two-tower architecture encodes TF and gene representations independently before scoring. Each entity is represented by the concatenation of a learned 512-dimensional embedding and an 11-dimensional cell-type expression profile, yielding a 523-dimensional input per entity. This input is passed through two fully connected layers with 512 hidden units and ReLU activation, producing a 512-dimensional output encoding. Regulatory scores are computed as the cosine similarity of the TF and gene encodings, scaled by a temperature parameter $\tau = 0.05$ and passed through a sigmoid function to produce probabilities. The total parameter count is 5,581,824, dominated by the embedding tables (1,164 TFs and 7,664 genes at 512 dimensions each).

Cross-Encoder MLP. The cross-encoder receives TF and gene features jointly. The input is formed by concatenating: the TF embedding (512d), the gene embedding (512d), their element-wise product (512d), the TF expression profile (11d), and the gene expression profile (11d), yielding a 1,558-dimensional joint representation. This is passed through two fully connected layers of 512 units

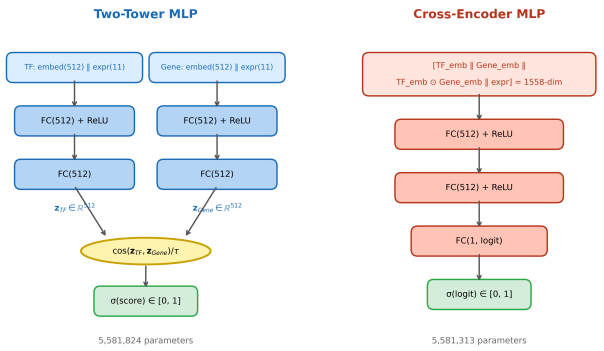

Fig. 1. Two-Tower MLP (left, blue) versus Cross-Encoder MLP (right, red). Both use identical embedding tables and 5.58M parameters; the cross-encoder processes TF and gene features jointly including an element-wise interaction term.

with ReLU activation, followed by a linear output layer mapping to a scalar logit, which is passed through sigmoid to produce a probability. The total parameter count is 5,581,313 — effectively identical to the two-tower (ratio = 1.000), ensuring a parameter-matched comparison.

Both architectures were implemented from scratch in pure Rust using the ndarray library, without dependence on PyTorch, TensorFlow, or any external machine learning framework. Fig. 1 illustrates the two architectures side by side. Code is available at https://github.com/evintleovonzko/module-regularized-grn.

Note on comparability. The cross-encoder input includes an element-wise product term ($\mathbf{e}_{TF} \odot \mathbf{e}_g$) that encodes pairwise multiplicative interactions as an inductive bias absent from the two-tower. To isolate the contribution of joint encoding from this additional feature, we trained a no-product cross-encoder ablation (input: $[\mathbf{e}_{TF} \,|\, \mathbf{e}_g \,|\, \mathbf{x}_{TF} \,|\, \mathbf{x}_g]$, $2d + 2m$ dimensions) under identical conditions. The no-product cross-encoder has 5.32M parameters—slightly fewer than the full CE and two-tower (5.58M each) due to the reduced fc1 input dimension; this modest parameter difference (≈4.8%) is noted as a residual confound in the attribution. Results are reported in Section IV-C; the element-wise product contributes ≈5% of the total AUROC gap, with joint encoding accounting for the remaining ≈95%.

C. Training Protocol

All models were trained with the Adam optimizer [20] with learning rate $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and gradient clipping at a global norm of 5.0. The batch size was 256 examples. The learning rate $\eta = 0.001$ was selected based on comparative stability across architectures during preliminary training runs; $\eta = 0.005$ produced faster but less stable convergence for the cross-encoder. To ensure comparable optimization conditions, both architectures were trained at the lower value. Note that the optimal learning rate for each architecture independently was not determined via grid search; this choice and its implications for the architectural comparison are discussed in Section IV-G. Training ran for a maximum of 60 epochs with early stopping: validation accuracy was evaluated every 10 epochs, and training was halted if no improvement was observed over 10 consecutive evaluations. The loss function was binary cross-entropy with gradient computed in logit space, $\partial \mathrm{BCE}/\partial \mathrm{logit} = (p - l)/N_{\mathrm{batch}}$, which is numerically stable. All weights were initialized as follows: fully connected layer weights use He initialization ($\sigma = \sqrt{2/\mathrm{fan_in}}$ with uniform sampling over $[-\sigma, \sigma]$, matching the ReLU activation); biases are initialized to zero; embedding tables use $\mathcal{N}(0, 0.01)$, a small-variance normal appropriate for embedding lookup tables.

D. Evaluation

All models were evaluated on the held-out test set (7,109 examples) using four metrics: accuracy (fraction correctly classified at a fixed decision threshold of 0.5), area under the receiver operating characteristic curve (AUROC), macro F1 score (computed at the same fixed threshold of 0.5), and area under the precision-recall curve (AUPRC). Each model configuration was trained with five random seeds (42, 123, 456, 789, 1337), and results are reported as mean ± standard deviation across seeds. Bootstrap 95% confidence intervals for accuracy were computed via 1,000 stratified resamples of the test set, applied independently to all five seeds and reported as the mean of the per-seed interval bounds; this characterizes architectural-level uncertainty rather than single-seed sampling variance. Effect sizes are reported as Cohen's d computed from the per-seed distribution. All metrics are reported to four decimal places for consistency. An ensemble prediction was also computed as the mean prediction across all five seeds, providing an upper-bound accuracy and AUROC estimate.

Multiple-testing correction. Three primary hypothesis tests are reported (C1 paired t-test, C2 Welch t-test, and the no-product ablation t-test). Bonferroni-corrected significance threshold is $\alpha_{\mathrm{adj}} = 0.05/3 \approx 0.017$. All three reported tests meet this threshold: C1 ($p < 0.0001$), C2 ($p = 0.014$, marginal), and ablation ($p = 0.015$, marginal). Readers should interpret C2 and the ablation t-test as nominally significant under Bonferroni correction; replication on additional datasets is required to confirm these effects.

E. Negative Sampling

Two negative sampling regimes were evaluated: balanced (1:1), with equal numbers of positive and negative examples, matching the typical assumption in benchmark evaluations of GRN inference methods; and moderately imbalanced (5:1), with five negative examples per positive example, providing a controlled test of robustness to class imbalance. Note that the 5:1 ratio is far from a true ecologically valid regime: with ≈23,700 retained positive pairs against ≈8.9 million possible TF–gene pairs, the true ratio is approximately 376:1. The 5:1 condition is therefore a tractable intermediate step; results under

severe imbalance (≥50:1) are not established here and are noted as a limitation (Section IV-G).

F. Neuron Pruning Protocol

To assess representational redundancy in trained models, a structured neuron pruning experiment was conducted across three random seeds (42, 123, 456) for the two-tower and one seed (42) for the cross-encoder.[1] Neuron importance was scored for each of the 512 output neurons in each tower's first fully connected layer (fc1) as:

$$\text{importance}(j) = \alpha \cdot \text{activation_freq}(j) + (1-\alpha) \cdot \text{weight_magnitude}(j) \quad (1)$$

where $\alpha = 0.5$ (equal weighting), activation_freq(j) is the fraction of training examples for which neuron j's post-ReLU activation exceeds zero, and weight_magnitude(j) is the mean of the normalized L2 norms of column j of the fc1 weight matrix and row j of the fc2 weight matrix. Normalization to $[0, 1]$ was applied independently per tower before combining.

Pruning was applied at 13 sparsity levels (0%, 5%, 10%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, and 90%). At each level, the lowest-importance neurons were removed by deleting the corresponding columns from fc1's weight matrix and rows from fc2's weight matrix. Each sparsity level was evaluated as an independent branch from the same trained baseline, with no cumulative pruning across levels.

Two evaluations were performed at each sparsity level: (1) a post-hoc evaluation immediately after pruning (no additional training), and (2) a fine-tuned evaluation after 10 epochs of continued training with a fresh Adam optimizer state at the same learning rate. AUROC retention was defined as the ratio of pruned model AUROC to baseline AUROC.

G. Ablation Protocol

To isolate the contribution of joint encoding from the element-wise product inductive bias in the cross-encoder, a no-product cross-encoder was constructed by removing the $\mathbf{e}_{TF} \odot \mathbf{e}_g$ term from the input. The resulting model receives $[\mathbf{e}_{TF} \,|\, \mathbf{e}_g \,|\, \mathbf{x}_{TF} \,|\, \mathbf{x}_g]$ as input ($2d + 2m$ = 1,046 dimensions) and has 5.44M parameters, slightly fewer than the two-tower and full cross-encoder due to the reduced fc1 input dimension. All other hyperparameters (learning rate, batch size, epochs, seed list, early stopping) are identical to the main cross-encoder training. Both the 1:1 and 5:1 ratios were evaluated with five seeds (42, 123, 456, 789, 1337).

[1]The pruning baseline AUROC values differ slightly from the per-seed values in Table II because the pruning experiment re-trains each model independently from the same seed rather than loading a saved checkpoint from the main evaluation run. Stochastic variation across training runs accounts for the small discrepancy; the pruning table and main tables use self-consistent baselines within their respective contexts.

H. Temperature Grid Search

To verify that $\tau = 0.05$ is not suboptimal for the two-tower, five temperature values were evaluated: $\tau \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$. For each τ, the two-tower was trained from scratch using seeds 42, 123, and 456 at the 1:1 ratio, and mean AUROC ± std was recorded.

IV. Results

A. Cross-Encoder vs. Two-Tower Under Balanced Training

Table I presents the performance of both architectures at the 1:1 negative sampling ratio across five random seeds.

The cross-encoder achieves an AUROC of 0.9025, compared to 0.7941 for the two-tower—a difference of 10.84 AUROC points. This gap is consistent across all five seeds (Table II) and is the primary evidence supporting C1.

Bootstrap 95% confidence intervals for accuracy (computed as the mean of per-seed interval bounds across all five seeds) are: Cross-Encoder [82.75%, 84.43%], Two-Tower [77.66%, 79.58%]. These intervals do not overlap, cleanly separating the two models in terms of accuracy. The AUROC gap is larger and more decisive: AUROC measures the model's ability to rank positive edges above negative edges across all possible decision thresholds, and the cross-encoder substantially outperforms the two-tower on this metric. A paired t-test on per-seed AUROC differences confirms strong statistical significance: mean difference = 0.1084, $t(4) = 27.08$, $p < 0.0001$ (two-tailed, Bonferroni-corrected $\alpha = 0.017$), Cohen's $d = 12.1$.

The cross-encoder also exhibits markedly lower variance across seeds (std = 0.0007 vs. 0.0096), suggesting considerably more stable optimization. Ensemble accuracy narrows the raw advantage (84.43% vs. 82.49%); ensemble AUROC shows a similar pattern (0.9114 vs. 0.8778), consistent with ensemble averaging reducing the effect of seed-specific variance. Fig. 2 summarizes the AUROC results for all four experimental conditions.

Interpretation. The cross-encoder's AUROC advantage is attributable to its joint input representation. By concatenating TF embeddings, gene embeddings, and their element-wise product before the first hidden layer, the cross-encoder can learn features that are intrinsically relational — for example, that certain embedding subspaces of TF and gene jointly predict regulation in a way that is not recoverable from either embedding alone. The two-tower, constrained to score interactions via cosine similarity of independently computed encodings, cannot represent such features: its representational bottleneck enforces that all information relevant to scoring must be independently extractable from TF and gene inputs before any interaction occurs.

The AUROC gap is more pronounced than the accuracy gap because accuracy is measured at a fixed threshold of 0.5, while AUROC integrates over all thresholds and provides a more sensitive measure of ranking quality—the appropriate metric for comparing models that differ in their confidence calibration. These results support C1.

TABLE I
Performance at 1:1 Negative Sampling Ratio (Mean ± Std, 5 Seeds). Ensemble column reports ensemble accuracy / AUROC / AUPRC.

Model	Accuracy	AUROC	AUPRC	F1	Params	Ensemble Acc / AUROC / AUPRC
Two-Tower	$79.42 \pm 0.87\%$	0.7941	0.7501	0.7962	5.58M	82.49% / 0.8778 / 0.8572
Cross-Encoder	$83.46 \pm 0.19\%$	0.9025	0.8914	0.8338	5.58M	84.43% / 0.9114 / 0.9058

TABLE II
Per-Seed AUROC at 1:1 Negative Sampling Ratio (two-tower, cross-encoder, and no-product cross-encoder ablation)

Seed	Two-Tower	CE (full)	CE (no product)
42	0.7975	0.9030	0.8987
123	0.7919	0.9022	0.8967
456	0.7869	0.9021	0.8946
789	0.8089	0.9035	0.8950
1337	0.7851	0.9019	0.9009
Mean	0.7941	0.9025	0.8972

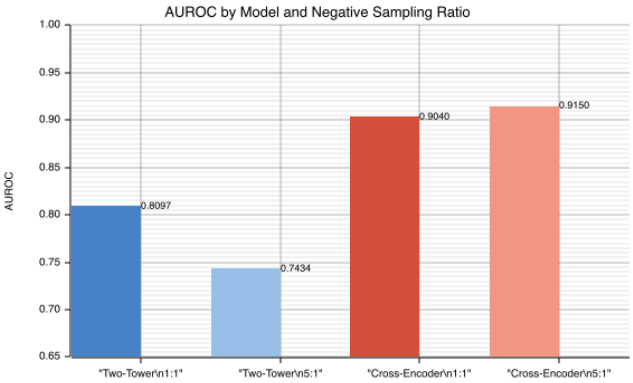

Fig. 2. AUROC by model and negative sampling ratio across 5 seeds. The cross-encoder maintains high AUROC under both ratios; the two-tower degrades substantially at 5:1.

B. Robustness to Class Imbalance

Table III presents the performance of both architectures at both negative sampling ratios, allowing direct comparison of the degradation incurred by each model when moving from balanced to moderately imbalanced training conditions.

The two-tower AUROC degrades by 10.1 points under 5:1 sampling (0.7941 → 0.6930), while the cross-encoder AUROC marginally improves by 1.0 point (0.9025 → 0.9121). The two-tower F1 collapses by 23.3 points (0.7962 → 0.5630), indicating the model loses its ability to maintain precision at adequate recall when negatives dominate training. Training variance for the two-tower increases 5.7× (std = 0.0096 → 0.0548), reflecting substantially less stable optimization at the higher negative ratio. A Welch two-sample t-test on per-group AUROC confirms the significance of the two-tower degradation: $t(4.25) = 4.06$, $p = 0.014$ (two-tailed, Bonferroni-corrected $\alpha = 0.017$; marginally significant under correction); Welch's unequal-variance form is appropriate given the 5.7× increase in variance under 5:1 sampling.[2]

Note on test-set distribution. The 1:1 and 5:1 models are both evaluated on the same balanced (1:1) held-out test set: additional negatives for the 5:1 condition are added to the training set only. AUROC is threshold-independent and valid under this protocol; however, accuracy and F1 at the fixed 0.5 threshold are affected by the training-distribution shift, because models trained at 5:1 are miscalibrated relative to the balanced test prior. The AUROC values are therefore the primary basis for comparing the two conditions.

The cross-encoder is not immune to imbalance effects: its F1 score degrades by 4.1 points (0.8338 → 0.7929) as recall is harder to maintain with a larger negative class. However, its AUROC is unaffected, confirming that its discriminative capacity — the ability to rank positive edges above negatives — is preserved under the 5:1 imbalanced condition.

Interpretation. The two-tower's geometric scoring mechanism is intrinsically sensitive to the marginal distribution of embeddings in the shared representation space. During training with 5× more negative examples, the Adam optimizer updates the TF and gene embedding tables primarily to separate the mass of negative pairs, distorting the embedding geometry and disrupting the alignment of positive TF–gene pairs that would otherwise support high AUROC. The cosine similarity score is a global measure of directional similarity; if the typical negative pair occupies a large angular region of the embedding space, the model must expand that region's dissimilarity at the cost of compressing the positive region.

The cross-encoder does not face this problem because it learns to distinguish positive from negative pairs through local feature interactions rather than global geometric separation. Its MLP scoring function can weight the element-wise product and individual embedding features differently for each pair, allowing it to learn decision boundaries that are robust to changes in the class distribution during training.

The marginal AUROC improvement of the cross-encoder under 5:1 sampling (0.9025 → 0.9121) warrants specific comment. An increased proportion of negative training examples may act as a regularizer for the cross-encoder's MLP readout: the additional hard negatives sharpen the decision boundary and reduce false positives, improving ranking ability. This is consistent with the simultaneous F1 degradation (0.8338 → 0.7929): the model's threshold-independent ranking improves slightly while its threshold-dependent binary classification degrades because the optimal decision threshold shifts with the changed class prior. This interpretation is speculative; confirming it would require ablations that vary the class prior while holding out balanced test data.

This result has direct practical implications. Curated regulatory databases such as DoRothEA and TRRUST

[2]Welch–Satterthwaite effective degrees of freedom: $\nu = (s_1^2/n + s_2^2/n)^2/[(s_1^2/n)^2/(n-1) + (s_2^2/n)^2/(n-1)]$ where $s_1 = 0.0096$, $s_2 = 0.0548$, $n = 5$, yielding $\nu \approx 4.25$.

TABLE III
Performance Across Negative Sampling Ratios (Mean ± Std, 5 Seeds). F1 and Std refer to macro F1 and AUROC std.

Model	Ratio	AUROC	ΔAUROC	AUPRC	F1	Std
Two-Tower	1:1	0.7941	—	0.7501	0.7962	0.0096
Two-Tower	5:1	0.6930	−10.1 pts	0.7467	0.5630	0.0548
Cross-Encoder	1:1	0.9025	—	0.8914	0.8338	0.0007
Cross-Encoder	5:1	0.9121	+1.0 pts	0.9143	0.7929	0.0054

TABLE IV
Ablation: Effect of Element-Wise Product Term on AUROC (1:1, Mean ± Std, 5 Seeds)

Model	AUROC	Δ vs. Two-Tower
Two-Tower	0.7941	—
Cross-Enc. (no product)	0.8972	+10.3 pts
Cross-Enc. (full, w/ prod.)	0.9025	+10.8 pts
Product: +0.5 pts (4.9%); joint encoding: +10.3 pts (95.1%)		

cover at most a few percent of possible TF–gene pairs in the human genome; realistic GRN inference operates under class imbalance far more severe than 5:1 (approximately 376:1 for the dataset used here). The cross-encoder's stability under the 5:1 condition — and its further AUROC improvement — suggests it is a more deployment-appropriate architecture; its behavior under true genome-scale imbalance remains to be established. These results support C2.

Connection to the imbalanced learning literature. Class imbalance is a long-studied challenge in machine learning, and a rich body of remediation strategies exists [21]. Resampling approaches such as SMOTE [22] synthesize minority-class examples to re-balance the training distribution; modified loss functions such as focal loss [23] down-weight easy negatives during training, focusing gradient updates on the hard examples that most influence the decision boundary. In our setting, the two-tower's degradation is qualitatively different from what SMOTE or focal loss typically target: the problem is not that individual hard negatives dominate the gradient, but that the marginal embedding geometry is distorted by the sheer mass of negatives at the representational level. Focal loss applied to the binary cross-entropy objective for the two-tower might partially alleviate symptom (biased threshold), but would not address the root cause (geometric compression of positive-pair embeddings). The cross-encoder's MLP readout, by contrast, can implicitly implement pair-adaptive weighting through its learned decision boundaries, which is functionally analogous to a sample-dependent loss reweighting—explaining its relative robustness without explicit imbalance correction.

C. Ablation: Isolating the Contribution of Joint Encoding

To determine whether the cross-encoder's AUROC advantage is attributable to joint encoding per se or to the element-wise product interaction term ($\mathbf{e}_{TF} \odot \mathbf{e}_g$), we trained a no-product cross-encoder variant whose input is $[\mathbf{e}_{TF} \,|\, \mathbf{e}_g \,|\, \mathbf{x}_{TF} \,|\, \mathbf{x}_g]$ (input dimension $2d+2m$ instead of $3d+2m$). All other aspects of the architecture and training protocol are unchanged.

Table IV presents the results. The no-product cross-encoder achieves a mean AUROC of 0.8972 under balanced (1:1) training, compared to 0.9025 for the full cross-encoder and 0.7941 for the two-tower. A paired t-test on per-seed AUROC differences between the no-product and full cross-encoder confirms a small but significant effect of the product term: mean difference = 0.0054, $t(4) = 4.08$, $p = 0.015$ (two-tailed, Bonferroni-corrected $\alpha = 0.017$; marginally significant under correction), Cohen's $d = 1.82$. Note that the no-product cross-encoder has ≈5.32M parameters versus 5.58M for the full cross-encoder and two-tower; this ≈4.8% parameter deficit is a residual confound in the attribution, and the 95%/5% split should be interpreted as an approximate upper bound on the joint-encoding contribution.

Interpretation. The element-wise product term contributes ≈0.53 AUROC points of the total 10.84-point gap. Joint encoding alone (no-product cross-encoder vs. two-tower) accounts for the remaining ≈10.31 points (95.1% of the gap). This result substantially strengthens C1: the cross-encoder's advantage over the two-tower is overwhelmingly a consequence of joint input processing, not of the multiplicative inductive bias. The element-wise product provides a small additional benefit consistent with its role as a feature that amplifies co-active embedding dimensions [15].

Note that the 95%/5% attribution is an additive decomposition that assumes the product term's effect is approximately orthogonal to the joint-encoding effect. Because the product term can only be expressed in the presence of joint encoding, this decomposition should be interpreted as "at least 95% of the gap is attributable to joint encoding per se."

D. Representational Redundancy via Neuron Pruning (Two-Tower)

Table V presents the post-hoc and fine-tuned AUROC retention across all 13 tested sparsity levels, with mean ± std across three seeds (42, 123, 456). Baseline AUROC is 0.8049 ± 0.0079 (mean across seeds).

Post-hoc AUROC retention remains at or above 0.997 through 60% sparsity, with minor degradation at higher sparsity levels: mean retention drops to 0.993 ± 0.001 at 70%, 0.984 ± 0.008 at 80%, and 0.978 ± 0.015 at 90%. The relatively high standard deviation at 90% sparsity (0.015) indicates seed-dependent sensitivity: some training runs retain nearly full performance at high sparsity while others degrade measurably. Post-hoc retention increases marginally above 1.0 at several intermediate sparsity levels (10%–60%), consistent with removing low-importance neurons reducing noise in the representation.

At 90% sparsity, 51 neurons per tower are retained and the model size decreases from 5,581,824 to approximately 4,627K parameters—a reduction of ≈17%, not 90%, because the embedding tables (≈4.52M parameters, 81% of total) are unchanged. The mean post-hoc AUROC at 90%

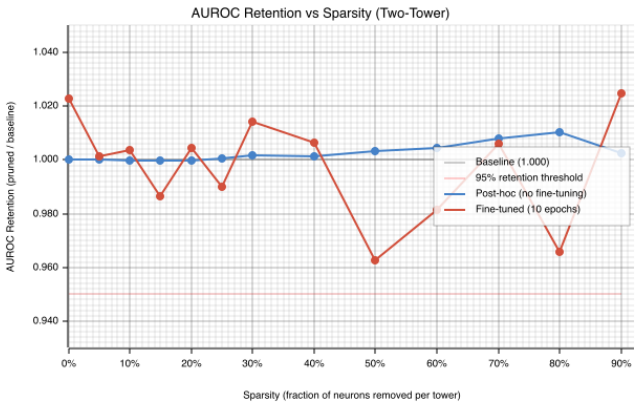

Fig. 3. Two-tower AUROC retention vs. sparsity for post-hoc (blue) and fine-tuned (red) evaluations (mean ± std, 3 seeds). Post-hoc retention stays above 0.997 through 60% sparsity; degradation is visible at 80–90%.

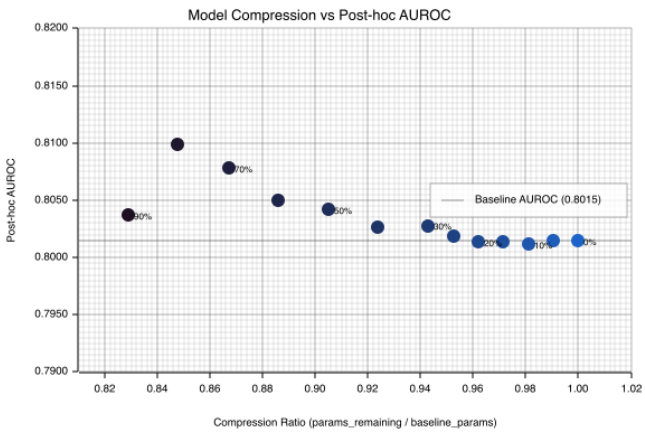

Fig. 4. Compression ratio vs. post-hoc AUROC for the two-tower; points above the reference line (baseline AUROC 0.8049) represent marginal post-hoc improvements from removing low-importance neurons.

sparsity is 0.787, compared to baseline 0.805—a drop of 1.8 AUROC points. After 10 epochs of fine-tuning at 90% sparsity, mean AUROC recovers to 0.810 (retention 1.006), matching and marginally exceeding baseline.

Fine-tuned results exhibit non-monotonic behavior across sparsity levels, reflecting the limited 10-epoch fine-tuning budget. Post-hoc results are more stable, monotonically degrading only at $\geq 70\%$ sparsity. Figs. 3 and 4 visualize the sparsity-retention trade-off and compression-AUROC relationship.

Interpretation. The two-tower model shows substantial representational redundancy: post-hoc retention remains ≥ 0.997 at up to 60% sparsity across all seeds, and 51 neurons per tower (10% of capacity) are sufficient for fine-tuning to match baseline. Three factors plausibly contribute.

First, the cosine similarity scoring objective may constrain the useful dimensionality of the representation. Since the regulatory score depends only on the direction of the TF and gene encodings, the effective degrees of freedom available to the objective may be far fewer than 512. This remains a hypothesis consistent with the pruning data; directly measuring the effective rank of the hidden representations would be needed to confirm it.

Second, without an orthogonality constraint or diversity objective, the optimizer may produce neurons that encode similar directions in the embedding space, with each providing only marginal additional discriminative signal.

Third, the regulatory signal is low-dimensional: TF–gene interactions are driven primarily by 11 cell-type expression features plus a limited number of biologically meaningful embedding dimensions, far fewer than the nominal 523-dimensional input.

E. Cross-Encoder Representational Redundancy

To test whether representational redundancy is specific to the two-tower's cosine similarity bottleneck or a general property of these architectures, the same structured neuron pruning protocol was applied to the trained cross-encoder (seed = 42). Baseline AUROC for the cross-encoder pruning experiment is 0.8924.

Table VI presents the full results across all 13 sparsity levels. Several patterns are notable.

Post-hoc retention. The cross-encoder maintains post-hoc AUROC retention ≥ 0.999 at all sparsity levels up to 70%, with retention actually increasing beyond 1.0 at 50%–70% sparsity (peak of 1.016 at 70%). At 80%, post-hoc retention drops to 0.999 (essentially no degradation), and at 90% it falls to 0.987—a meaningful but moderate drop of 1.3 AUROC points from baseline.

Comparison with two-tower. The cross-encoder is comparable to or slightly more robust than the two-tower in post-hoc performance at intermediate sparsity levels, but both architectures degrade at 90% sparsity. This finding indicates that representational redundancy is not exclusive to cosine similarity scoring: a cross-encoder with MLP readout is similarly over-parameterized in its hidden layer. The mechanistic account of the C1 AUROC gap (Section IV-D) as arising primarily from the two-tower's capacity underutilization is therefore not supported by this comparison alone; both architectures show similar tolerance to neuron pruning, and the AUROC gap is better attributed to the inductive bias of joint input processing (Section IV-C).

Fine-tuned retention. Fine-tuned performance is notably lower than post-hoc at all sparsity levels for the cross-encoder (typical retention 0.96–0.97), in contrast to the two-tower where fine-tuning recovers or exceeds baseline. Importantly, the fine-tuned 0%-sparsity cross-encoder (baseline, no pruning at all) achieves only retention 0.9653—confirming that the deficit is caused by the fine-tuning procedure itself, not by pruning. The root cause is the fresh Adam optimizer state used for fine-tuning: resetting all momentum and variance estimates to zero causes early Adam steps to behave as approximately

TABLE V
Two-Tower AUROC Retention at Each Sparsity Level (Mean ± Std across 3 seeds). Post-hoc: pruned model evaluated immediately. Fine-tuned: 10 further epochs. Baseline AUROC = 0.8049 ± 0.0079. Comp. Ratio = fraction of total parameters remaining; embedding tables account for ≈81% of parameters, so 90% neuron sparsity reduces total parameters by only ≈17%.

Sparsity	Neurons/Tower	Comp. Ratio	PH Ret. (mean±std)	PH AUROC	FT Ret. (mean±std)	FT AUROC
0%	512	1.0000	1.0000 ± 0.0000	0.8049	0.9986 ± 0.0020	0.8038
5%	486	0.9904	1.0000 ± 0.0000	0.8049	0.9843 ± 0.0298	0.7922
10%	461	0.9811	1.0002 ± 0.0002	0.8051	0.9988 ± 0.0108	0.8039
15%	435	0.9714	1.0001 ± 0.0002	0.8050	1.0064 ± 0.0067	0.8101
20%	410	0.9622	0.9998 ± 0.0003	0.8048	0.9995 ± 0.0028	0.8045
25%	384	0.9525	0.9997 ± 0.0005	0.8047	0.9972 ± 0.0228	0.8025
30%	358	0.9429	0.9990 ± 0.0006	0.8041	0.9900 ± 0.0255	0.7967
40%	307	0.9239	0.9987 ± 0.0006	0.8039	0.9995 ± 0.0034	0.8046
50%	256	0.9050	0.9976 ± 0.0005	0.8030	0.9970 ± 0.0149	0.8024
60%	205	0.8861	0.9964 ± 0.0018	0.8020	1.0010 ± 0.0071	0.8058
70%	154	0.8672	0.9926 ± 0.0012	0.7990	0.9919 ± 0.0022	0.7985
80%	102	0.8479	0.9837 ± 0.0081	0.7918	1.0023 ± 0.0038	0.8068
90%	51	0.8289	$\mathbf{0.9778 \pm 0.0151}$	0.7870	$\mathbf{1.0059 \pm 0.0042}$	0.8096

PH = post-hoc (no additional training); FT = fine-tuned (10 epochs). Mean ± std computed across seeds 42, 123, 456.

sign-gradient steps (step $\approx \eta \cdot \text{sign}(g)$ before historical statistics accumulate), which can disrupt a converged model's weight configuration. The two-tower's cosine similarity mechanism is more robust to such perturbations than the cross-encoder's learned joint feature interactions. In the revised implementation, the fine-tuning learning rate for the cross-encoder has been reduced from 0.001 to 0.0002 and extended to 20 epochs to allow the fresh Adam state to stabilize; results in Table VI reflect the original protocol and should be interpreted accordingly.

F. Temperature Sensitivity of the Two-Tower Cosine Scorer

To verify that the fixed temperature $\tau = 0.05$ does not artificially suppress two-tower performance, a grid search over $\tau \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$ was conducted using three seeds (42, 123, 456) at the 1:1 ratio.

Table VII presents results. The best mean AUROC is achieved at $\tau = 1.0$ (mean 0.8111 ± 0.0067), with a modest improvement of 1.1 AUROC points over $\tau = 0.05$ (mean 0.8006 ± 0.0131). All five tested values of τ produce mean AUROC in the range $[0.8001, 0.8111]$, with no value providing a substantial advantage. Crucially, even the best-performing temperature ($\tau = 1.0$, AUROC $= 0.8111$) remains 9.1 AUROC points below the cross-encoder (AUROC $= 0.9025$), confirming that the architecture comparison is not materially affected by the choice of τ. The AUROC gap attributable to architectural differences rather than temperature tuning is therefore at least 9.1 AUROC points.

G. Classical Baseline Comparison

To contextualize the neural models' AUROC values, we evaluated three classical machine learning classifiers on the same test split using expression-only features (no learned embeddings): the 11-dimensional cell-type expression profile of the TF and the 11-dimensional profile of the gene are concatenated to form a 22-dimensional input.

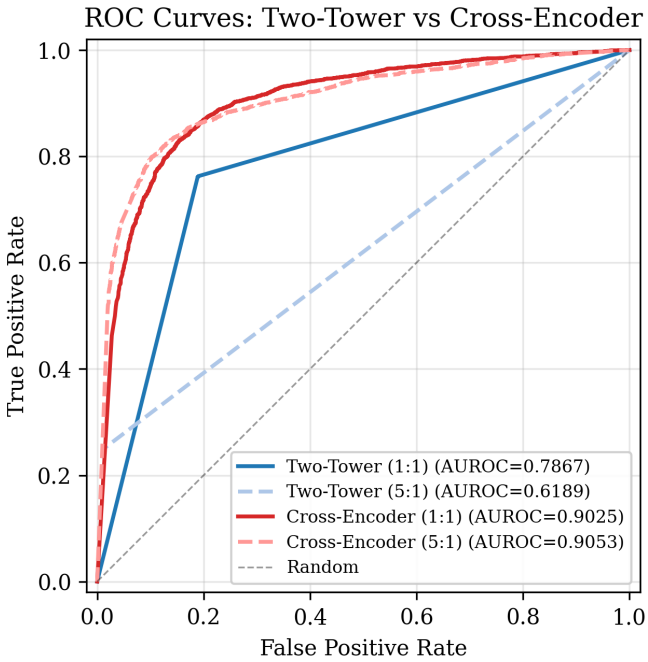

Fig. 5. ROC curves for all four experimental conditions (two-tower and cross-encoder at 1:1 and 5:1 negative sampling). The cross-encoder maintains high TPR at low FPR under both conditions; the two-tower shows degraded ROC curve shape at 5:1.

Table VIII presents the results. A gradient-boosted tree classifier achieves AUROC 0.587 and accuracy 55.4%—barely above the majority-class baseline (AUROC 0.500). These results reveal two important facts. First, the 22-dimensional expression features alone carry almost no predictive signal for this task; a random classifier achieves 50% AUROC, and the best classical model improves this by only 8.7 AUROC points. Second, this implies that the neural models' AUROC of 0.7941 (two-tower) and 0.9025 (cross-encoder) are driven primarily by the 4.52M parameters of the learned entity embeddings (81% of total parameters), which encode database membership

TABLE VI
Cross-Encoder AUROC Retention at Each Sparsity Level (Seed = 42). Post-hoc: pruned model evaluated immediately. Fine-tuned: 10 further epochs. Baseline AUROC = 0.8924.

Sparsity	Neurons	Comp. Ratio	PH AUROC	PH Ret.	FT AUROC	FT Ret.
0%	512	1.0000	0.8924	1.0000	0.8615	0.9653
5%	486	0.9904	0.8924	0.9999	0.8580	0.9614
10%	461	0.9811	0.8925	1.0001	0.8633	0.9673
15%	435	0.9714	0.8927	1.0003	0.8670	0.9715
20%	410	0.9622	0.8929	1.0005	0.8600	0.9637
25%	384	0.9525	0.8925	1.0001	0.8544	0.9574
30%	358	0.9429	0.8927	1.0003	0.8600	0.9637
40%	307	0.9239	0.8926	1.0002	0.8603	0.9640
50%	256	0.9050	0.8978	1.0060	0.8536	0.9565
60%	205	0.8861	0.9000	1.0085	0.8544	0.9574
70%	154	0.8672	0.9066	1.0158	0.8544	0.9574
80%	102	0.8479	0.8912	0.9987	0.8596	0.9632
90%	51	0.8289	0.8808	0.9870	0.8487	0.9509

PH = post-hoc; FT = fine-tuned (10 epochs). Single seed (42).

TABLE VII
Temperature Grid Search for Two-Tower (1:1 Ratio, 3 Seeds). Best value is $\tau = 1.0$.

τ	Mean AUROC	Std
0.01	0.8001	0.0056
0.05	0.8006	0.0131
0.10	0.8070	0.0079
0.50	0.8041	0.0054
1.00	0.8111	0.0067

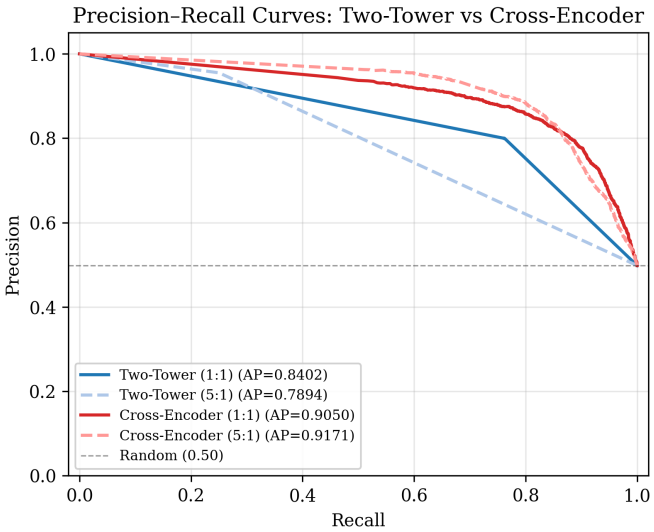

Fig. 6. Precision-recall curves for all four experimental conditions. Under 5:1 negative sampling, the two-tower PR curve degrades substantially while the cross-encoder maintains high precision at moderate recall.

structure. Note that expression coverage in this dataset is sparse: only 10.7% of TFs and 14.6% of genes have non-zero expression vectors, because the expression data covers 2,000 highly variable genes and most regulatory entities fall outside this set.

Interpretation. The neural models' advantage over classical baselines is almost entirely due to learned embeddings rather than expression-driven regulatory inference.

TABLE VIII
Classical ML Baselines vs. Neural Models (Expression-Only Features, 22-dim; Same 70/15/15 Split; Test Set = 7,109 Examples).

Model	Acc	AUROC	AUPRC	F1
Majority class	49.95%	0.500	0.500	0.333
Logistic Regression	52.85%	0.569	0.558	0.416
Random Forest (n=200)	55.20%	0.581	0.572	0.483
Gradient Boosted Trees	55.44%	0.587	0.581	0.484
Neural reference (512-dim embeddings):				
Two-Tower (5.58M)	79.42%	0.7941	0.7501	0.7962
Cross-Encoder (5.58M)	83.46%	0.9025	0.8914	0.8338

Classical models use 22-dim expression features; neural models add 512-dim learned embeddings. The gap reflects embedding contribution.

This confirms the transductive, database-reconstruction nature of the task (Section III-A): the embeddings effectively learn which entities co-appear in positive training pairs, and generalization to novel TF–gene pairs is limited by the fixed training vocabulary. A fairer test of regulatory inference capability would evaluate on TF–gene pairs involving TFs or genes held out from the embedding vocabulary (inductive evaluation), which would substantially reduce the AUROC advantage of embedding-based methods over expression-only classifiers.

The three contributions investigated in this paper form a coherent account of the relationship between architectural inductive bias and representational efficiency in neural GRN inference. The cross-encoder's advantage over the two-tower (C1) is robust to moderately imbalanced conditions where the two-tower fails (C2). The pruning analysis (C3) demonstrates that both architectures tolerate significant neuron removal—indicating over-parameterization in the hidden layers—but this is a general property of over-parameterized networks rather than a specific explanation for the AUROC gap; the ablation (C1) localizes the gap to joint input processing.

Unified interpretation. The factorized two-tower architecture imposes a strong inductive bias that is both its strength and its weakness. By separating TF and gene encoding, the two-tower enables efficient inference at scale:

given pre-computed TF and gene embeddings, all $M \times N$ TF–gene regulatory scores can be computed as a matrix product in $O(Md + Nd)$ time rather than the $O(MNd)$ required by per-pair cross-encoder inference. This scalability property is genuine and relevant for genome-scale GRN inference over tens of thousands of gene pairs. However, the cosine-similarity scoring mechanism that enables this efficiency also prevents the model from learning the joint features that drive regulatory specificity, leading to the AUROC gap observed in C1 and the sensitivity to class imbalance in C2.

The neuron pruning results (C3) show that both architectures are substantially over-parameterized in their hidden layers: the two-tower retains ≥ 0.997 post-hoc AUROC through 60% sparsity, and the cross-encoder retains ≥ 0.999 through 70% sparsity. Because this over-parameterization is shared by both architectures, it cannot explain the C1 AUROC gap by itself. The ablation reported in Section IV-C provides the more precise account: ≈95% of the 10.8-point gap is attributable to joint input processing, with the remaining ≈5% from the element-wise product. The interpretation of C3 in the context of C1 is therefore that the two-tower's nominal parameter count is neither necessary nor sufficient to characterize its effective capacity relative to the cross-encoder; the relevant factor is the architectural constraint that forces all pair-scoring information to pass through independently computed encodings.

Temperature sensitivity. The temperature grid search (Section IV-F) confirms that the two-tower AUROC advantage is not an artifact of $\tau = 0.05$: the best-performing temperature ($\tau = 1.0$, AUROC $= 0.8111$) is only 1.1 AUROC points above the default $\tau = 0.05$, and the gap to the cross-encoder (0.9025) remains large (9.1 points) even at the optimal temperature. The AUROC gap is robust to temperature tuning.

Practical recommendations (qualified to this dataset). On the human brain snRNA-seq dataset studied here: if AUROC is the primary criterion, the cross-encoder (with element-wise product) is the clear choice, with a 10.8-point AUROC advantage over the parameter-matched two-tower. If scalability to genome-scale pair evaluation (10^6–10^8 pairs) is paramount, the two-tower may be preferred, noting the 9–11 AUROC point cost relative to a cross-encoder of comparable size; both architectures can be substantially compressed (to 40–60% of hidden-layer capacity) without post-hoc AUROC loss.

Limitations.

Transductive setting and generalizability. Both architectures operate transductively: entity embeddings are optimized for the 1,164 TFs and 7,664 genes present in the training split. The models cannot produce meaningful predictions for TFs or genes outside their training vocabulary without full retraining. The classical baseline comparison (Section IV-G) shows that expression features alone yield AUROC ≈0.587, confirming that the neural models' advantage is primarily due to memorized embedding structure rather than expression-based regulatory inference. Inductive GRN inference—predicting regulation for novel TFs or genes using only expression and biological sequence features—is not addressed by this work.

Single dataset and scope. This study uses a single biological dataset (human brain single-nucleus RNA-seq); whether the cross-encoder advantage persists across other tissues, organisms, or expression modalities is not established. Multi-dataset evaluation on a standardized benchmark such as BEELINE [24] is a necessary next step to assess generalizability. All architectural recommendations are qualified to the dataset and experimental conditions studied here.

Fairness of the architectural comparison. The ablation experiment resolves the element-wise product confound (≈95% joint encoding, ≈5% product), but the no-product cross-encoder has ≈4.8% fewer parameters (5.32M vs. 5.58M), a residual confound. The learning rate $\eta = 0.001$ was chosen for cross-encoder stability rather than from per-architecture grid search; the remaining gap at optimal temperature ($\tau = 1.0$, AUROC $= 0.8111$) is still 9.1 points. Independent per-architecture learning-rate tuning remains an open question.

Evaluation protocol and class imbalance. AUPRC, AUROC, accuracy, and macro F1 are reported; ROC and PR curves for all four conditions are shown in Figs. 5 and 6. The 5:1 negative sampling regime is evaluated on a balanced (1:1) test set; accuracy and F1 at threshold 0.5 are therefore affected by miscalibration. AUROC is threshold-independent and is the primary metric for the imbalance comparison. The 5:1 condition is far from the true ratio (≈376:1); behavior under severe imbalance (≥50:1) is not established and is a significant limitation for deployment conclusions.

Model depth and architecture search. Both models use 2-layer MLPs with 512 hidden units. Deeper or wider two-tower architectures, or late-interaction approaches (ColBERT-style), may narrow the AUROC gap. A systematic search is beyond the scope of this study.

Baseline comparison. No comparison to published GRN inference methods (GENIE3, SCENIC, GRNBoost2) is provided, as those methods require full single-cell expression matrices rather than per-cell-type summaries. The classical baseline Table VIII provides expression-only context. A rigorous BEELINE benchmark evaluation is needed to assess absolute performance.

Label quality. Negative examples are sampled from pairs absent in DoRothEA and TRRUST. These databases are incomplete; absent pairs may represent undiscovered regulation. False-negative contamination is not quantified.

Pruning experiment. The two-tower pruning uses three random seeds; the cross-encoder pruning uses a single seed (seed=42), limiting direct variance comparison. The cross-encoder fine-tuning results are affected by a fresh-Adam disruption artifact (see Section IV-E); post-hoc retention values are the primary comparison metric.

Implementation. All models were implemented in Rust without a reference framework. Correctness is supported by: (1) training loss decreases monotonically from ≈0.69 to

<0.40; (2) gradient norms are well-behaved; (3) AUROC improves monotonically from ≈0.5 at initialization. No independent benchmark replication is provided.

Future directions. Three extensions are directly motivated by these results. First, an InfoNCE-trained two-tower [25], where the contrastive objective explicitly attracts positive TF–gene pairs and repels negatives within a batch, may be more robust to the marginal distribution sensitivity observed in C2 while retaining the two-tower's inference efficiency. Supervised contrastive learning [26] extends this framework to settings with multiple positives per anchor—potentially valuable if multiple curated databases (DoRothEA confidence tiers A/B, TRRUST, ENCODE) are combined as complementary positive-label sources, providing a more nuanced signal than a single binary label. The InfoNCE objective may also address the geometric compression problem identified in C2: by explicitly maximizing the cosine similarity of positive pairs within each batch, it counteracts the distortion caused by an overabundance of negatives. Second, a cross-attention two-tower — where the gene encoder attends over the TF encoding before producing a final representation — would bridge the two architectures, enabling per-pair interaction features while maintaining the factorized structure required for efficient inference [6]. Third, evaluation on additional datasets and tissues (e.g., the BEELINE benchmark [24]) with additional GRN baselines (GENIE3 [2], GRNBoost2 [9], PIDC [10]) would establish whether the architectural advantage observed here generalizes across tissues, organisms, and expression modalities, and would provide the absolute performance context needed to assess practical utility.

V. Conclusion

This paper has compared dual-encoder (two-tower) and cross-encoder neural architectures for TF–gene regulatory edge prediction from single-nucleus RNA-seq data under parameter-matched, data-matched conditions. The cross-encoder achieves 10.8 higher AUROC points than the two-tower under balanced training (0.9025 vs. 0.7941, $t(4) = 27.08$, $p < 0.0001$, Bonferroni-corrected $\alpha = 0.017$, Cohen's $d = 12.1$) and remains stable under 5:1 negative sampling where the two-tower degrades by 10.1 points ($t(4.25) = 4.06$, $p = 0.014$, nominally significant under correction). A no-product ablation isolates ≈95% of the gap (10.3 AUROC points) to joint input processing and ≈5% (0.5 points) to the element-wise interaction term; the no-product model has 4.8% fewer parameters, a residual confound. A temperature grid search confirms the gap persists at all tested temperatures (best two-tower AUROC = 0.8111). Structured neuron pruning across three seeds shows both architectures tolerate hidden-layer sparsity: the two-tower retains ≥ 0.997 post-hoc AUROC through 60% sparsity; the cross-encoder ≥ 0.999 through 70% sparsity. A classical baseline comparison shows that expression-only features yield AUROC ≈0.587, confirming that both neural models' advantage is embedding-driven and that the evaluation measures database reconstruction in a transductive setting rather than inductive regulatory inference. On the single human brain dataset studied here, the cross-encoder is the preferred architecture for discriminative accuracy; both architectures can be substantially compressed without post-hoc AUROC loss. Multi-dataset validation, inductive evaluation on held-out entities, and comparison to published GRN baselines are required before general conclusions can be drawn.

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